

PAIND description

Matteo Frongillo

- **Title:** Cold-water circulation with a heat pump
- **Supervisor:** Ludger Fischer
- **Industrial partner / CC:** R. Nussbaum AG / Patrik Zeiter
- **Description:**

The project consists of a domestic water system with separate domestic cold water (DCW) and domestic hot water (DHW) distribution. The DHW side includes a circulation loop to maintain hygiene and allow disinfection against Legionella. The key idea is to create a closed DCW loop before the domestic distribution and use it as the circuit for a heat pump for heat extraction and DHW preparation.

This configuration keeps the domestic distribution hygienically safe and assesses whether the cold-water loop can reliably support the heat pump for the production of domestic hot water.

The scope is to size the main hydraulic and thermal components and test the technical feasibility of the heat pump, including the evaporator, condenser, compressor, expansion valve, circulation pumps, and a possible storage connection.